

DPCN

Assignment 1

Opinion Network Formation

Team - Elenjical Fanbois

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Github link to codebase: https://github.com/sohanwithanm/DPCN_A1

1 Preparing the data

The dataset contains responses to a survey designed to capture opinions on four broad domains: Technology, Education, Society and Ethics, and Environment. Each domain contains 15 statements, resulting in 60 opinion statements in total. The original dataset contains 96 response records and one additional column containing an anonymized response identifier. The response identifier was retained only for internal record tracking and was not used as a numerical feature during network construction.

Table 1: Structure of the survey dataset

Prefix	Domain	Number of statements
T	Technology	15
E	Education	15
S	Society and Ethics	15
V	Environment	15
Total		60

1.1 Validation of the Raw Data

Before constructing either network, the dataset was checked for structural and categorical consistency. The following validation checks were performed:

- The dataset contained exactly 60 survey-question columns.
- Each domain contained the expected 15 questions.
- No duplicate response identifiers were present.
- Every observed response belonged to one of the five expected Likert categories or was recorded as **No Comments**.
- Blank cells were distinguished from observed Likert responses.

These checks ensured that unexpected response labels or incorrectly formatted question columns were not silently included in the numerical analysis.

1.2 Likert-Scale Encoding

The original responses were categorical and ordinal. They were converted to a numerical scale from 1 to 5, as shown in Table 2.

Table 2: Numerical encoding of the Likert-scale responses

Response	Numerical value
Strongly Disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly Agree	5

The transformation preserves the ordering of the response categories. No reverse coding was applied because the purpose of the analysis was to examine relationships among the statements as they were presented, rather than to combine them into a single composite attitude score. Consequently, a negative association between two questions remains substantively meaningful and may indicate contrasting response tendencies.

1.3 Treatment of Missing Responses

Blank entries and **No Comments** responses were treated as missing values and represented as NaN.

The raw dataset contained 505 blank cells and 37 **No Comments** entries. Together, these accounted for 542 of the 5,760 possible survey responses, corresponding to an initial missingness rate of approximately 9.41%.

Five submissions contained no answers to any of the 60 questions. These records provided no information for either network and were removed before analysis. After removing them, 91 respondents remained. The retained data matrix contained 242 missing entries out of 5,460 possible responses, corresponding to a missingness rate of approximately 4.43%.

Table 3: Summary of dataset completeness

Quantity	Value
Raw submissions	96
Completely empty submissions removed	5
Respondents retained for analysis	91
Completely answered submissions	68
Survey questions	60
Missing entries before removing empty submissions	542
Missing entries among retained submissions	242
Missingness among retained submissions	4.43%

1.4 Construction of the Analysis Matrix

After validation, encoding, and removal of the five empty submissions, the data were represented as a matrix

$$\mathbf{X} \in \mathbb{R}^{91 \times 60}, \quad (1)$$

where each row represents a respondent and each column represents a survey question. An observed entry x_{ij} contains the encoded Likert response given by respondent i to question j , while an unanswered item remains missing.

The same matrix was used to construct both networks, but with different orientations:

- For the respondent-based network, each row of $\mathbf{X}$ represents the opinion profile of one respondent across the survey questions.
- For the question-question network, each column of $\mathbf{X}$ represents the response pattern of one question across respondents.

2 Network Formation

2.1 Respondent based Network

- The network was constructed using each respondent as a distinct node.
- Edges were established using a pairwise-complete Pearson correlation, which calculates the ideological distance between pairs based only on the specific survey items both individuals answered.
- To ensure correlations were structurally sound, a minimum overlap floor of 10 co-answered items was required to correlate any pair. The observed overlap ranged from 12 to 60 items, with 55.6% of the 4,095 pairs sharing all 60; no pair fell below the floor.
- The similarity r_{ij} between respondent i and respondent j over the set S_{ij} of items that both answered was calculated as:

$$r_{ij} = \frac{\sum_{k \in S_{ij}} (v_{ik} - \bar{v}_i)(v_{jk} - \bar{v}_j)}{\sqrt{\sum_{k \in S_{ij}} (v_{ik} - \bar{v}_i)^2} \sqrt{\sum_{k \in S_{ij}} (v_{jk} - \bar{v}_j)^2}}$$

Here S_{ij} is the set of items both respondents *answered*; whether their answers agreed is what r_{ij} then measures. Centring each vector on this shared subset ensures that uniform agreement and uniform disagreement are treated as opposed rather than similar.

- An undirected edge was retained when $|r_{ij}| \geq \tau$, with $\tau = 0.40$. Thresholding the *absolute* correlation keeps strongly opposed respondents as negative edges rather than discarding them; each edge stores the signed value of r_{ij} , mirroring the treatment of ρ_{ab} in the question-question network.
- After thresholding and removing 3 isolated respondents (ids 119, 120, 123), the final analysis graph consisted of **88 nodes and 970 ties: 957 positive and 13 negative**.

The choice of τ is the single free parameter of this construction, so it is worth showing what it selects. The left panel of Figure 1 gives the distribution of all 4,095 pairwise correlations, which range from -0.676 to $+0.774$ with a mean of $+0.252$. A threshold of 0.40 sits at the 77th percentile and retains the strongest 23.4% of pairs as edges. Note also that 11.0% of pairs are *negatively* correlated, representing genuinely opposed opinion profiles; because the threshold is positive, these never become edges, so the network represents agreement only.

The right panel shows the similarity matrix with respondents grouped by their eventual Louvain community. A network with strong community structure would show bright blocks along the diagonal against a pale background. Here the field is almost uniformly positive with only faint diagonal blocks, which is the visual counterpart of the low modularity reported in Table 5.

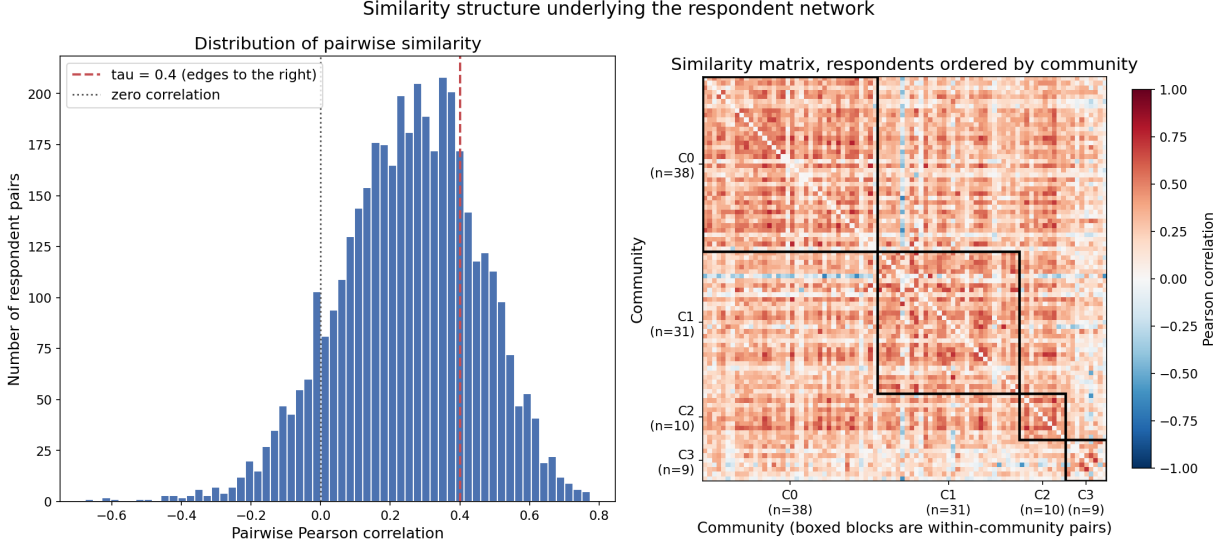


Figure 1: Similarity structure underlying the respondent network. Left: distribution of pairwise Pearson correlations, with the edge threshold $\tau = 0.40$ marked. Right: the similarity matrix with respondents ordered by community; the weak block structure on the diagonal corresponds to the Louvain partition.

2.2 Question - Question Opinion Network

The second network was constructed to examine relationships between the survey statements themselves. Unlike the respondent-based network, where nodes represent individuals, the nodes in this network represent the 60 survey questions. An edge indicates that two questions exhibited sufficiently similar or contrasting response patterns across the surveyed respondents.

- Each of the 60 survey questions was represented as a node, with its corresponding domain - Technology, Education, Society and Ethics, or Environment - stored as a node attribute.
- An edge between two questions represented an association between their response patterns across respondents.
- Pairwise-complete Spearman correlation was used because the Likert responses are ordinal. Only respondents who answered both questions were included, with a minimum requirement of 60 shared responses. The actual pairwise sample sizes ranged from 81 to 91.
- The association between questions a and b was represented by

$$\rho_{ab} = \text{corr}(\text{rank}(Q_a), \text{rank}(Q_b)).$$

- An undirected edge was retained when

$$|\rho_{ab}| \geq 0.30.$$

The edge weight was $w_{ab} = |\rho_{ab}|$, while the sign indicated whether the relationship was positive or negative.

- The final network contained 60 nodes and 328 edges, of which 322 were positive and 6 were negative. It contained one main component of 57 questions and three isolated questions.
- Positive edges were subsequently used for Louvain community detection, while negative edges were retained for identifying contrasting opinion patterns.

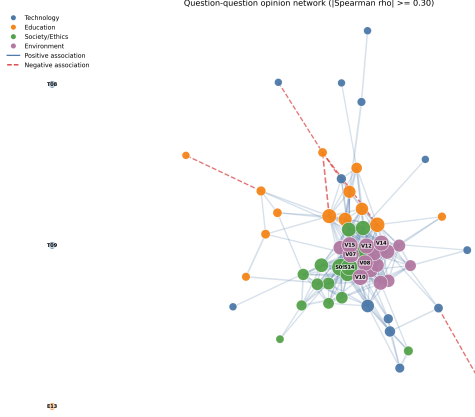


Figure 2: The visualization represents questions as nodes, coloured according to their survey domain. Node size indicates weighted connectivity, while edge thickness represents correlation strength. Blue edges show positive associations and dashed red edges show negative associations. Only the most strongly connected questions are labelled to maintain readability.

This network therefore captures the empirical structure of relationships among the surveyed beliefs. While the original survey divided the statements into four predefined domains, the question - question network allows the observed response patterns to determine whether those domains remain distinct or combine into broader cross-domain themes.

3 Analysis and visualizations

3.1 Respondent based Network

3.1.1 Global structure

The network forms a single connected component containing all 88 respondents. Its global properties are summarised in Table 4.

Table 4: Global structural metrics of the respondent network ($|r_{ij}| \geq \tau$, $\tau = 0.40$)

Metric	Value
Nodes (respondents)	88
Edges (positive 957, negative 13)	970
Density	0.253
Mean degree	22.05
Median degree	20.5
Minimum / maximum degree	1 / 51
Average clustering coefficient	0.540
Transitivity	0.590
Connected components	1
Diameter	5
Mean shortest path length	1.97
Degree assortativity	+0.057

About one in four of all possible respondent pairs holds opinion profiles correlating at $|r_{ij}| \geq \tau$. The average clustering coefficient of 0.540 is more than twice the 0.260 produced by an Erdős-Rényi random

graph of identical size, indicating that agreement is strongly transitive: if two respondents both agree with a third, they usually agree with each other. Combined with a mean shortest path of 1.97, this is the signature of a cohesive rather than fragmented opinion space. Degree assortativity is close to zero, so respondents holding mainstream views show no tendency to agree preferentially with one another; there is no in-group effect.

Degree structure of the opinion network ($\tau = 0.4$)

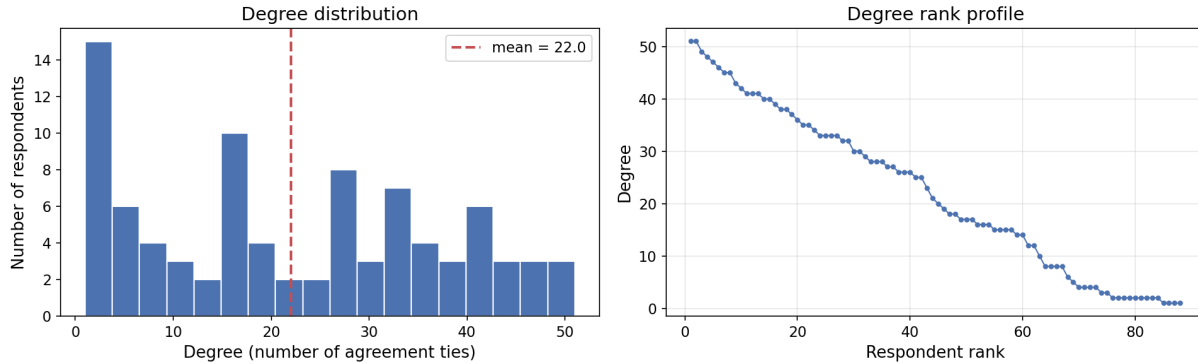


Figure 3: Degree structure of the respondent network. The left panel shows the degree distribution; the right panel shows the degree-rank profile, in which respondents are ordered from most to least connected. The smooth, near-linear decline indicates a continuous gradient of mainstream-ness rather than a small set of dominant hubs.

Because degree here counts how many classmates share a respondent’s opinion pattern, the degree-rank profile in Figure 3 is effectively a ranking of how mainstream each respondent is. The ten most connected respondents account for only about a quarter of all tie endpoints, far below what a hub-dominated network would show, while a tail of respondents holds only one or two ties and represents genuinely idiosyncratic opinion profiles.

In an opinion network the interpretation of centrality differs from a social network. Degree centrality identifies respondents whose views are most widely shared, marking the mainstream of the class rather than a popular individual; respondents 49 and 61 lead with 51 ties each. Betweenness centrality instead identifies respondents sitting on the shortest paths between otherwise distant viewpoints, and therefore acts as an ideological bridge.

Degree, closeness and eigenvector centrality are almost perfectly correlated with one another, whereas betweenness correlates with degree only moderately. This separation identifies the substantively interesting cases: respondents who bridge far more than their connectivity would predict. Respondent 87 ranks 66th of 88 by degree yet 5th by betweenness, and respondent 26 ranks 68th by degree but 14th by betweenness. These are minority voices whose opinion profiles nevertheless blend positions that are elsewhere separated.

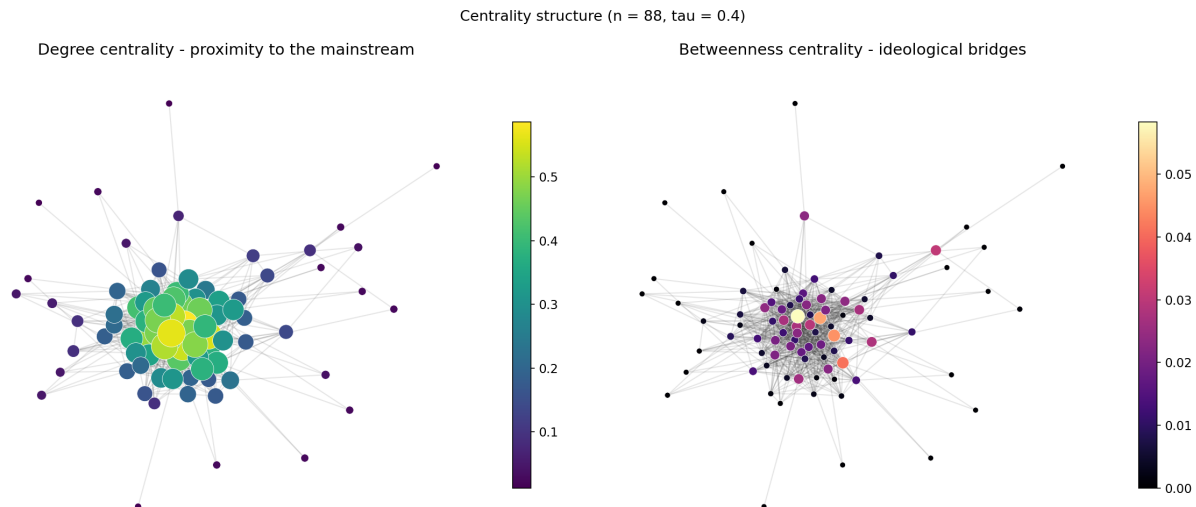


Figure 4: The respondent network drawn with node size proportional to degree. Colour shows degree centrality in the left panel and betweenness centrality in the right panel. Nodes that are bright in the right panel but small in the left are ideological bridges.

3.1.2 Community detection and null-model benchmarking

Because the Louvain algorithm cannot optimise positive and negative weights simultaneously, the partition was computed on the positive-edge subgraph (88 nodes, 957 edges), as in the question-question network. It returns four communities of sizes 38, 31, 10 and 9.

Modularity must be benchmarked before a partition can be interpreted, since Louvain returns one whether or not community structure exists. Louvain is also sensitive to the order in which nodes are presented, so a single run would overstate precision. Modularity was therefore computed across 20 node orderings and compared against two null models: an Erdős-Rényi graph fixing node and edge counts, and a configuration model additionally preserving the degree sequence.

Table 5: Modularity of the positive-edge subgraph against null models

Model	Modularity Q	z -score
Observed (20 node orderings)	0.1532 ± 0.0053	—
Erdős-Rényi null	0.1451 ± 0.0049	+1.66
Configuration-model null	0.1431 ± 0.0042	+2.42

Modularity sits modestly above both nulls, and is detectable against the stricter degree-preserving comparison ($z = +2.42$). It nevertheless remains far below the value of approximately 0.3 conventionally required to claim well-separated communities.

The partition itself, however, is not reproducible. Re-ordering the nodes of the *same* graph under the same random seed yields between four and five communities whose sizes range from (38, 37, 8, 5) to (29, 27, 17, 15), with Q varying only between 0.149 and 0.158.

The community sizes quoted above are therefore illustrative rather than a stable property of the data. Such instability is the expected consequence of a modularity this close to the random baseline, and constitutes the clearest single piece of evidence that the class does not divide into opinion factions.

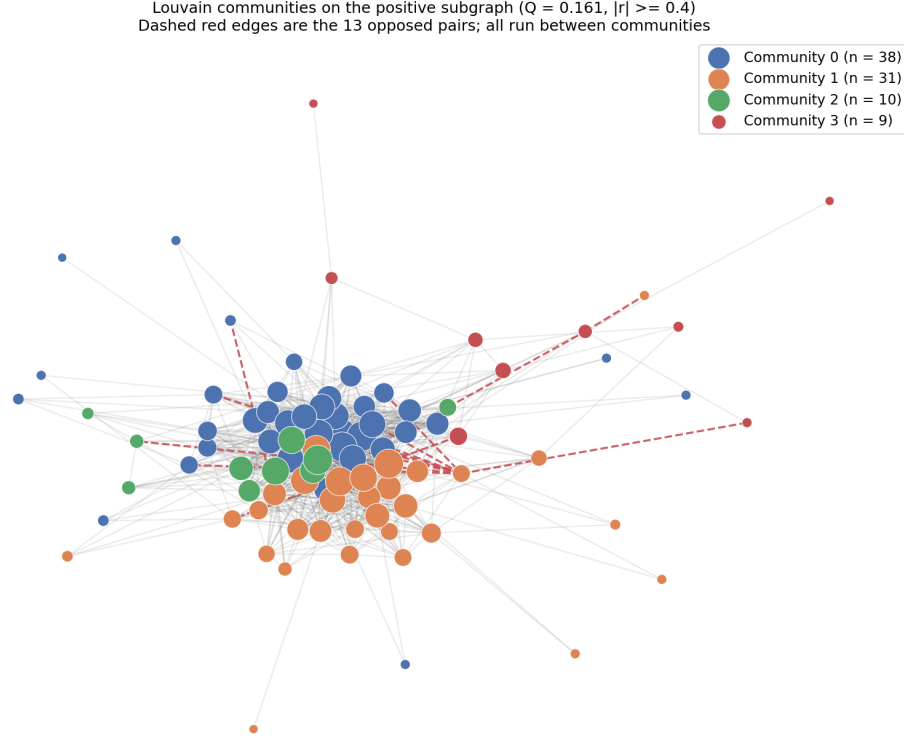


Figure 5: Louvain communities on the positive-edge subgraph. Dashed red edges mark the 13 opposed pairs. The heavy density of edges running between differently coloured groups illustrates why modularity is low.

3.1.3 Opposed respondents

Because edges were thresholded on $|r_{ij}|$, the network retains 13 negative edges (1.3% of all ties) joining respondents whose opinion profiles are strongly opposed. The strongest is $r = -0.676$ between respondents 30 and 82.

These edges provide an independent test of the partition. If the communities corresponded to genuine factions, opposition should run *between* them rather than within. This is exactly what is observed: **all 13 negative edges join different communities, and none falls inside one**. Given the community sizes, 4.2 within-community negative edges would be expected by chance, and the probability of observing none is $p = 0.006$.

The partition therefore separates genuine opposition, of which the class contains very little: only 13 of 970 ties are oppositional. The finding qualifies the broader picture of a consensual class rather than contradicting it, and rests on a small number of edges.

3.1.4 Threshold sensitivity

Every structural claim above is conditional on the similarity threshold τ . Because τ simultaneously governs density, connectivity and modularity, it was swept across a wide range; results appear in Table 6.

Table 6: Sensitivity of the respondent network to the similarity threshold (representative values; the full sweep steps by 0.05)

τ	Retained	Retained (%)	Edges	Density	Components	Modularity
0.20	91	100.0	2532	0.618	1	0.055
0.30	89	97.8	1745	0.446	1	0.090
0.40	88	96.7	957	0.250	1	0.155
0.55	62	68.1	230	0.122	3	0.289
0.70	21	23.1	20	0.095	4	0.586

Modularity rises monotonically with τ , from 0.055 at $\tau = 0.20$ to 0.586 at $\tau = 0.70$, which might appear to justify a stricter threshold. It correlates at -0.975 with the share of respondents retained. At $\tau = 0.65$ the network looks convincingly factional precisely because it has discarded 53% of the class and fragmented into three components. Most of the apparent community structure is therefore a property of the cut rather than of the students. The chosen value of $\tau = 0.40$ retains 96.7% of respondents in a single connected component while discarding the weakest three quarters of pairwise correlations.

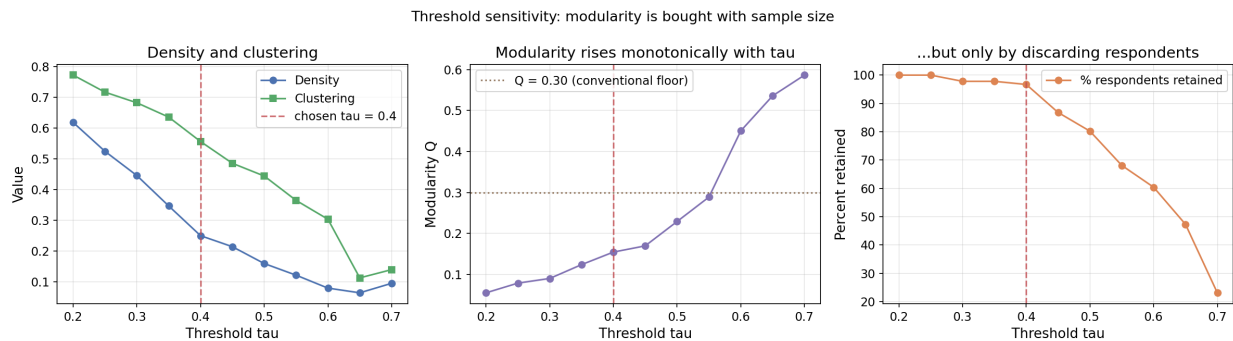


Figure 6: Threshold sensitivity of the respondent network. Modularity increases with τ (centre panel) but only at the cost of discarding respondents (right panel), the signature of a consensual rather than factional population.

3.1.5 Opinion profiles of the communities

To interpret the partition, each community was profiled across the four survey domains on the original 1–5 scale; the means are shown in the left panel of Figure 7.

Every community mean in every domain lies above the neutral midpoint of 3.0, the full range being 3.84 to 4.69, and the largest gap between any two communities in any domain is 0.40 points on a five-point scale. The communities therefore differ in the *intensity* of their agreement rather than its *direction*, which is precisely why modularity is low: a partition can only be strong if the groups it separates are genuinely different.

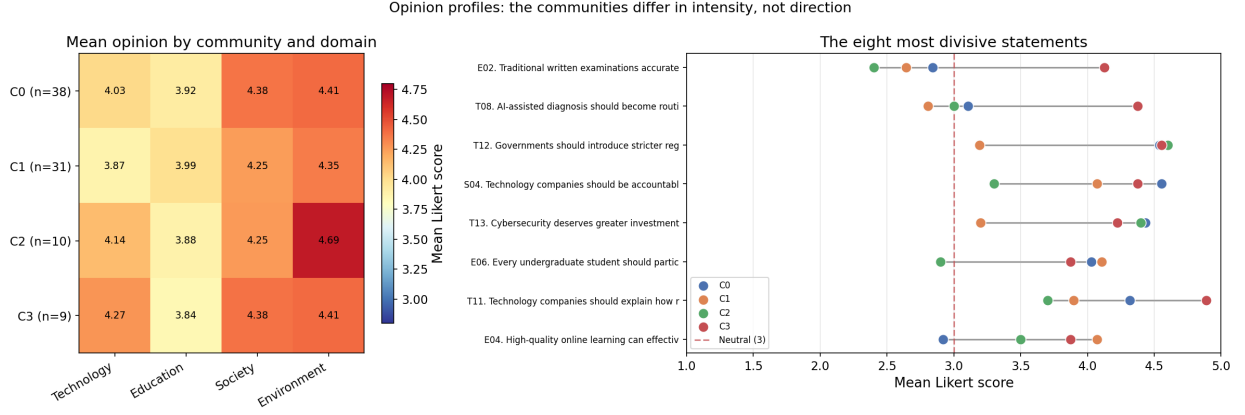


Figure 7: Left: mean opinion by community and domain. Right: the eight statements on which the communities differ most, with the neutral midpoint marked. The communities are separated by degree of agreement rather than by opposition.

3.2 Question - Question Opinion Network

At the primary threshold of $|\rho| \geq 0.30$, the question network contained 60 nodes and 328 edges. Of these, 322 were positive and 6 were negative. The network density was 0.185, with one main component containing 57 questions and three isolated questions.

Table 7: Sensitivity of the question network to the correlation threshold

Threshold	Edges	Density	Components	Isolates
0.25	494	0.279	1	0
0.30	328	0.185	4	3
0.35	189	0.107	13	12

The response-distribution visualization in Figure 8 shows that agreement was common across the survey, particularly for Society/Ethics and Environment questions. Education questions such as compulsory attendance and the accuracy of traditional examinations exhibited greater disagreement and variation.

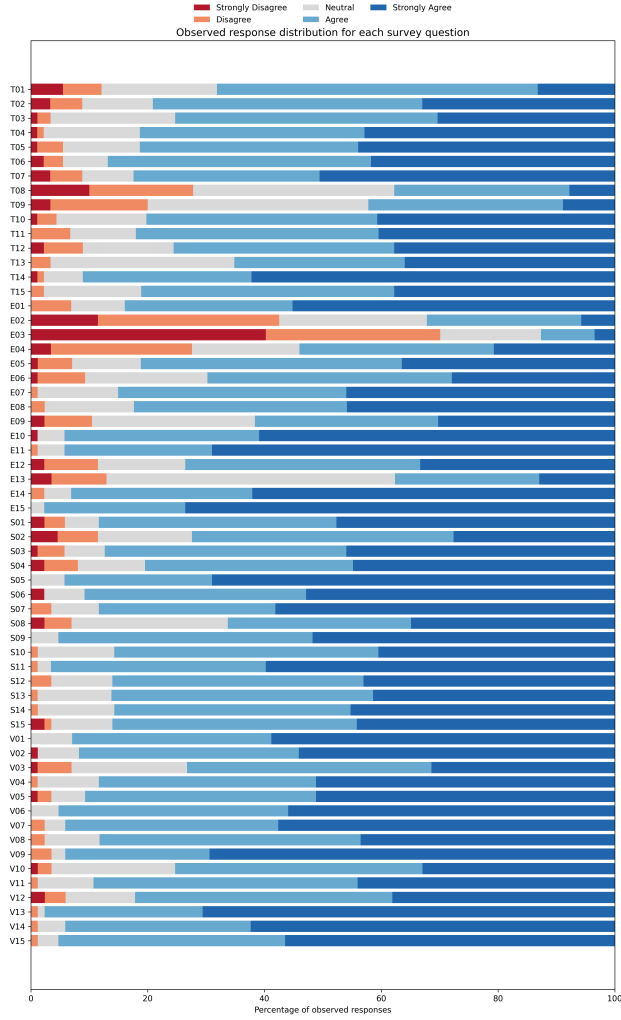


Figure 8: Distribution of observed Likert responses for the 60 survey questions.

Figure 9 presents the complete Spearman correlation matrix. The Environment block contains many strong positive correlations, while Technology has comparatively weaker internal associations. Cross-domain associations are also visible, particularly between Society/Ethics and Environment.

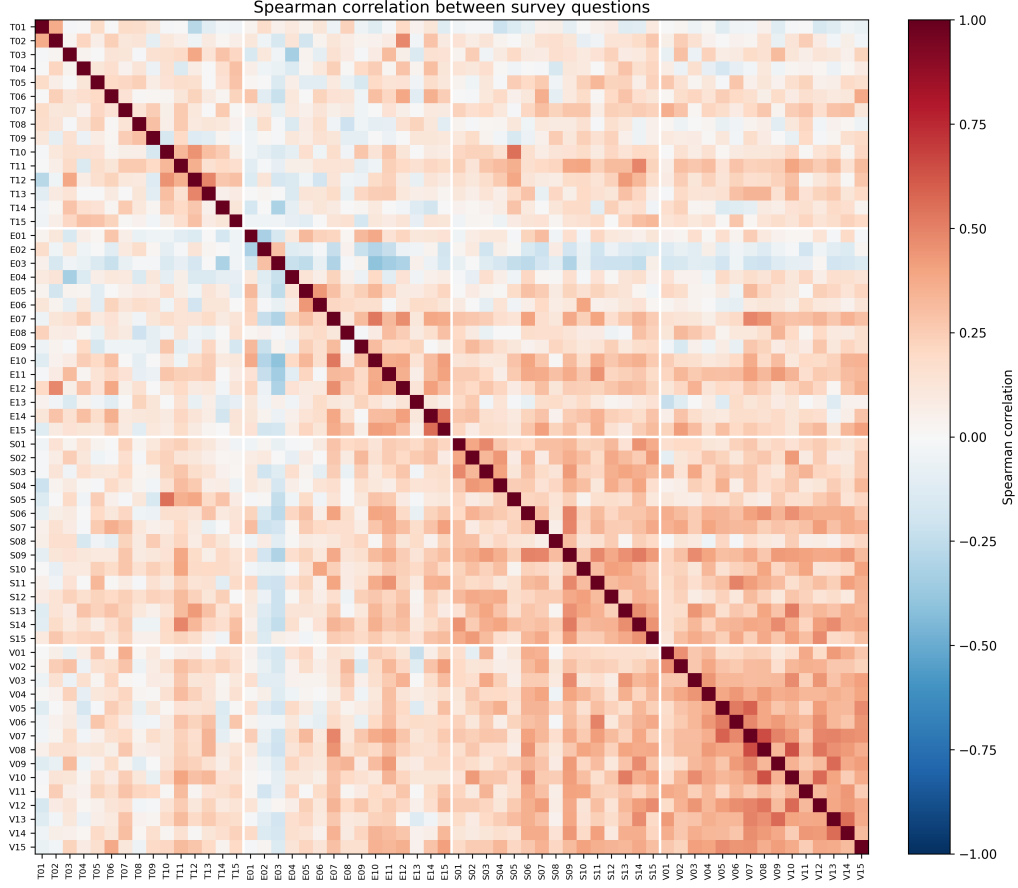


Figure 9: Pairwise-complete Spearman correlations between survey questions.

The strongest positive associations included biodiversity protection with sustainable campus practices ($\rho = 0.652$), sustainable campus practices with environmentally responsible consumption ($\rho = 0.633$), and biodiversity protection with restrictions on single-use plastics ($\rho = 0.591$). Strong cross-domain relationships included data protection with individual control over personal data ($\rho = 0.547$). The strongest negative relationship was between compulsory attendance and innovation-oriented education ($\rho = -0.405$).

Weighted node strength identified S09, S14, V07, V15, and V08 as the most strongly connected questions. These questions concern inclusion, ethical leadership, biodiversity, intergenerational responsibility, and sustainable university practices.

Figure 10 compares connectivity within and between the four survey domains. Environment exhibited the greatest internal connectivity, with 76.2% of possible within-domain pairs retained. This was followed by Society/Ethics at 47.6%, Education at 25.7%, and Technology at 6.7%. The strongest cross-domain mixing occurred between Society/Ethics and Environment, where 34.7% of possible pairs were retained.

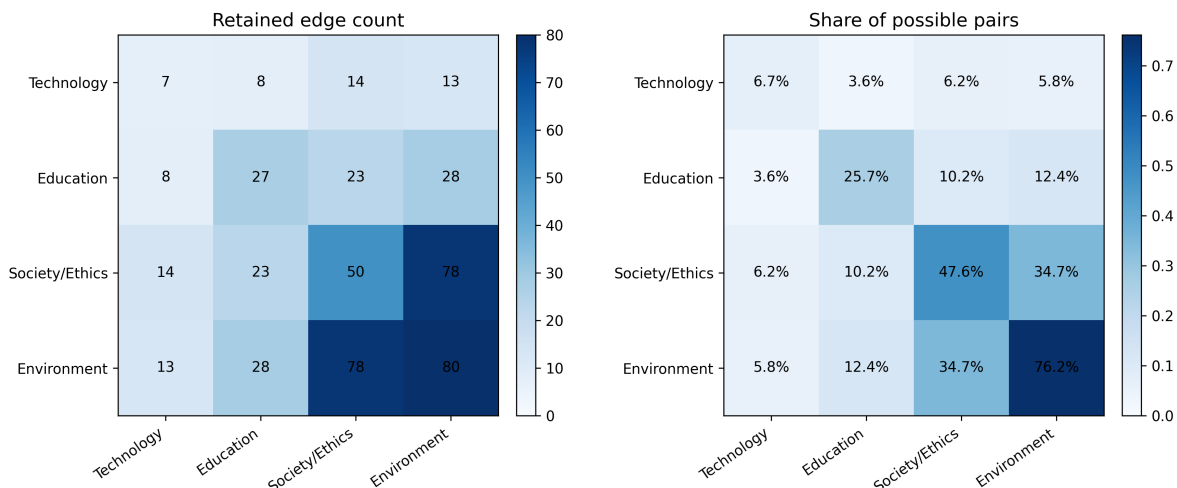


Figure 10: Number and proportion of retained edges within and between survey domains.

Louvain community detection was applied to the positive-edge subgraph. Four communities were detected among the positively connected questions, while seven questions had no positive edge above the selected threshold. The weighted modularity was 0.241, indicating a modest community structure rather than sharply separated groups.

Finally, the respondent-centred robustness network contained 41 edges, of which 25 were also present in the primary network. The Jaccard overlap was 0.073. This comparison was used to identify a smaller set of associations that remained after controlling for respondents' general tendency to agree.

4 Results and Discussion

4.1 Respondent based Network

- **The network reflects a consensual population, not a factional one.** All 88 respondents form a single connected component with high clustering (0.540 against 0.260 for a comparable random graph) and a mean shortest path of 1.97. Only 13 of 970 ties (1.3%) are oppositional.
- **Community structure is detectable but not reproducible.** Across 20 node orderings modularity averages $Q = 0.1532 \pm 0.0053$, exceeding a degree-preserving null by 2.42 standard deviations yet falling far below the conventional 0.3 benchmark. Critically, re-ordering the same graph yields four or five communities with substantially different sizes, so no specific partition is a stable property of the data.
- **Communities differ in intensity rather than direction.** No community-by-domain mean falls below the neutral midpoint, and the largest between-community gap in any domain is 0.40 points.
- **Opposition, where it exists, falls between communities.** All 13 negative edges join different communities and none falls within one, against 4.2 expected by chance ($p = 0.006$). The partition separates genuine opposition; there is simply very little of it.
- **The class agrees overwhelmingly.** Of 60 statements, only two are rejected outright: compulsory class attendance (mean 2.06, the strongest single opinion in the survey in either direction) and the claim that written examinations accurately measure knowledge (mean 2.84). The strongest consensus attaches to continuous learning (4.71), corporate environmental accountability (4.67) and individual control over personal data (4.63).
- **Disagreement concentrates in Technology and Education.** Of the fifteen most divisive statements, seven are Technology items, five Education and three Society; no Environment item appears at

all. The flashpoints are AI governance (T12, spread 1.51), cybersecurity investment (T13, 1.48), the accuracy of written examinations (E02, 1.27) and online learning (E04, 1.26).

- **Threshold dependence is the principal methodological caveat.** Modularity correlates at -0.975 with the share of respondents retained, so apparent factions sharpen only as the sample shrinks.

Both networks are built by thresholding a correlation matrix, but with coefficients and cut-offs calibrated to their own distributions: Pearson $|r| \geq 0.40$ here against Spearman $|\rho| \geq 0.30$ for the questions. Although the values differ, both sit near the 80th percentile of their respective correlation distributions (77th and 81st), producing comparable densities.

The overall structure of class opinion is therefore one of **shared values and contested implementation**: near-unanimous agreement on ends such as sustainability, privacy, openness and lifelong learning, with division over the technological and institutional means of achieving them.

4.2 Question - Question Opinion Network

Louvain detection identified four positive-edge communities with a modularity of 0.241, indicating modest rather than sharply separated community structure. Based on their dominant question content, these communities were interpreted as technology governance and privacy, educational innovation and AI adoption, ethics and social responsibility, and environmental sustainability. These are descriptive labels assigned by us, not outputs produced by the algorithm.

Environment was the most cohesive domain, retaining 76.2% of its possible internal edges. Strong connections were observed among biodiversity, sustainable practices, responsible consumption, and environmental accountability. Society/Ethics and Environment also showed the strongest cross-domain connection. In contrast, Technology retained only 6.7% of its possible internal edges, suggesting that technology opinions depended strongly on context, such as education, privacy, or public accountability.

The primary network contained 328 edges, while the respondent-centred robustness network contained only 41, with 25 edges shared between them. This indicates that general agreement tendency contributed to the density of the primary network. Nevertheless, several environmental, privacy, and educational relationships remained stable. Overall, the results suggest overlapping themes in class opinion rather than sharply separated ideological groups. Since the findings depend on the selected threshold and are correlational, they should not be interpreted as causal relationships.

5 Contribution

- **Soham Acharya (2024113012):** Led data preparation, including the numerical mapping of ordinal responses and the strict missing-data policy. Designed the edge formation logic using pairwise-complete Pearson correlation, constructed the Respondent-based Network, and managed the GitHub codebase.
- **Arjun Dingankar (2024113024):** Formulated the Question-Question Opinion Network. Calculated structural network metrics including density, clustering coefficients, and degree/betweenness centralities for both networks. Generated the threshold sensitivity and degree distribution visualizations.
- **Sumit Kumar (2022111012):** Implemented community detection via the Louvain algorithm and rendered the complex structural visualizations (community mapping and centrality graphs). Led the analysis for the Results and Discussion section, synthesized the final opinion profiles, and managed the L^AT_EX formatting of the report.